PIR Project - CITI Lab. - Chroma Team - I-Factory

Code Optimization and Simulation of a Self-Coordinating Robotic Aggregate

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Context: Designing robust robotic systems that can adapt timely to unknown environmental constraints at low energy and computation cost is a major challenge in robotics. In the most common robotics paradigm, a central computer processes information from a well defined set of sensors and communication devices to control a well defined set of actuators in displacement. In such a context, adaptability typically goes with more actuated degrees of freedom and sensing parameters, which is detrimental to robustness, as well as energy and computation efficiency. To address this problem, people have proposed more distributed design approaches, similar to biological systems [5], toward robotic matter [1].
A promising approach is to use principles of soft matter [2,3] to design robotic materials that rely on their complex mechanical properties to achieve tasks with minimal silicon computation. We proposed a granular robotic material design, the Granulobots (Fig. 1) as the first autonomous modular system that can deform like an active soft material, exhibiting rigid to soft solid and liquid response properties [4]. We showed that, when coupled with self-organizing properties, this allows a robotic aggregate made of multiple identical units to self-coordinate locomotion gait in different environmental conditions, and in a decentralized and mechanical way. This research contribute to a broader picture aiming toward a "co-design" framework, integrating well chosen mechanical properties with algorithmic development to optimize robustness, autonomy, as well as material, manufacturing and energy efficiency.

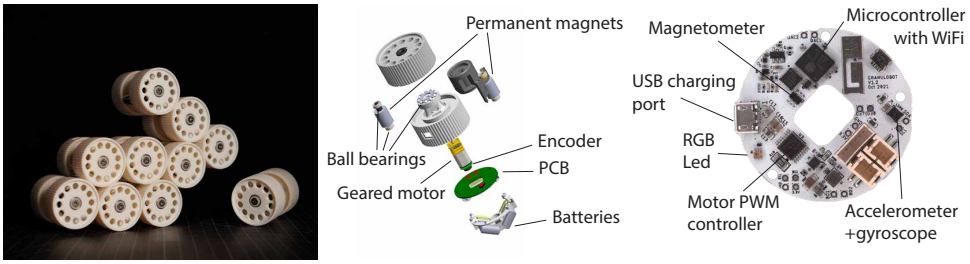


Figure 1: The Granulobot Platform.

Project: In this project we will explores strategies toward robotic aggregates that can choose autonomously the right parameters to self-coordinate their locomotion in a given environment. We will experiment directly on the hardware platform, and will explore the development of a simulation platform to simplify the testing protocols of distributed algorithms, and to extend it to a larger number of particles.

Project steps:

- We will start by presenting the Granulobot platform to the students, going through the mechanical design, the electronic circuit, the firmware code and software platform to communicate with the robot.
- One subgroup will work on optimizing the existing firmware code of the granulobot while the other group will develop a software simulator (we will first identify which existing simulator can be used to build a simulation).
- Both group will implement on their respective platforms, strategies to help robotic aggregate find self-coordinating locomotion states from arbitrary initial parameters.
- if time allow, both group will work together to compare the results between the hardware robots and the simulation platform.

References

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[3] T. Vicsek, A. Zafeiris. "Collective motion", *Physics Reports*, 2012.

[4] **B. Saintyves**, M. Spenko, H. Jaeger. "A self-organizing robotic aggregate using solid and liquid-like collective states", *Science Robotics*, 10.1126/scirobotics.adh4130, 2024

[5] **O. Simonin**, O. Grunder A cooperative multi-robot architecture for moving a paralyzed robot, *Mechatronics*, 2009